Qwen

Beyond Unfalsifiability: A Strategic Blueprint for Validating a Planck-Scale Quantum Gravity Framework Against Environmental and Thermal Challenges

Deconstructing the Unfalsifiability Claims: The Dual Challenge of Background Noise and Thermal Entanglement

The primary obstacle to the scientific validation of any proposed quantum gravity framework, particularly one centered on Planck-scale physics, is the formidable challenge of isolating a minute, theoretically predicted signal from a cacophony of environmental and intrinsic noise sources. The claim of unfalsifiability is not a dismissal of the theory's mathematical elegance but a pragmatic assessment of its ability to generate testable predictions within the constraints of current and foreseeable technology. This section undertakes a rigorous deconstruction of the two principal categories of blockers identified in the research goal: overwhelming background contributions and the profound complexities introduced by entanglement in thermal states. By dissecting the nature of these challenges, we establish a clear foundation for developing targeted strategies to mitigate them, thereby transforming perceived barriers into focal points for innovation. The central task is to demonstrate that while these issues are significant, they are not insurmountable and represent the most critical frontiers for both theoretical refinement and experimental ingenuity.

The first category of blocker, overwhelming background contributions, represents the most immediate and tangible threat to the detection of a quantum gravitational signature, conventionally denoted as ΔS_{cert}. In any laboratory setting, even those designed for extreme precision, signals are inevitably obscured by various forms of noise. For experiments probing the quantum nature of gravity, this problem is exacerbated by the expected weakness of the interaction at macroscopic scales. Thermal noise is a dominant and pervasive source of uncertainty, fundamentally limiting the sensitivity of instruments like interferometers [141]. The very act of making a measurement introduces uncertainties related to entropy and temperature, which are inherent to quantum theory

[1]. For instance, tabletop experiments aiming to detect gravity-induced entanglement using atom interferometry face severe challenges from decoherence. One analysis indicates that if the decoherence rate reaches 0.125 Hz, witnessing the entanglement between two qubits would necessitate encoding information in 6-dimensional qudits, a significant technical hurdle beyond current capabilities [40]. Similarly, proposals using table-top Michelson laser interferometers to probe spacetime fluctuations must contend with distinguishing a genuine quantum gravity signal from classical noise sources [108]. The Hannover thermal noise experiment highlights how thermal noise is a primary limitation for the sensitivity of gravitational wave detectors, a parallel challenge in precision measurement [141]. The problem is not merely one of detector quality but is rooted in the fundamental physics of measurement, where distinguishing a small effect from the statistical fluctuations of a noisy environment is a core difficulty in quantum metrology [12].

Beyond thermal noise, other background contributions include detector imperfections, electromagnetic interference, and vibrations. Any protocol must account for these systematic errors, which can mimic the signatures of novel physical phenomena, leading to false positives or, more dangerously, masking a true signal [58] [59]. The concept of "science spoofs" and parodies in physics underscores the historical difficulty of confidently attributing an anomalous signal to a new fundamental interaction rather than a subtle instrumental artifact [58] [60]. Therefore, a successful experimental strategy must incorporate robust methods for background characterization and subtraction. Research into characterizing thermal noise independently of quantum noise suggests that advanced data analysis techniques could potentially disentangle different noise components, providing a path toward cleaner signals [140]. Furthermore, theoretical models must go beyond idealized scenarios and explicitly calculate the expected magnitude of competing decoherence mechanisms—such as interactions with a squeezed thermal bath—which can produce distinct patterns of entanglement decay and decoherence [157][158]. Without a detailed quantitative comparison between the predicted quantum gravity signal (ΔS_{cert}) and the expected background noise, any claim of detection remains speculative.

The second, and perhaps more profound, category of blocker involves the complexities of entanglement when the system under study is not in a simple vacuum or product state but is instead interacting with a thermal bath or environment. This issue strikes at the heart of many foundational analyses in quantum foundations and quantum gravity, where the assumption of an unentangled initial state is often taken for granted [159]. However, in any realistic experimental scenario, the system of interest is part of a larger universe and will inevitably interact with its surroundings. When a system interacts with a thermal bath, it becomes entangled with that bath from the outset [2] [149]. The resulting

state is a mixed state, not a pure product state, and its properties are governed by the laws of statistical mechanics and open quantum systems [129]. The thermofield double state, a cornerstone of the AdS/CFT correspondence, provides a powerful mathematical description of a system in thermal equilibrium as a maximally entangled pure state across two copies of the system's Hilbert space [63] [64] [65]. Assuming a simple product state for such a system would be physically incorrect and could lead to erroneous conclusions about the generation of new correlations.

This complexity is compounded by the potential for the quantum channel mediating the gravitational interaction to be "entanglement breaking" [7]. An entanglement-breaking channel is a completely positive trace-preserving map that, when acting on one part of an entangled state, always produces a separable (non-entangled) final state. If the gravitational interaction, even if mediated by a quantum field, acts through such a channel, it would be incapable of generating entanglement between two distant systems. Instead, it might only induce local decoherence or other classical correlations. This presents a severe obstacle: even if the underlying theory of quantum gravity were correct, the observable effect might be the destruction of pre-existing entanglement rather than its generation, rendering it difficult to distinguish from other decoherence processes. The dynamics of entanglement in such scenarios are highly non-trivial, depending on the precise nature of the interaction Hamiltonian and the initial state of the environment [157] [158]. For example, a two-qubit system interacting with a squeezed thermal bath via a dissipative interaction exhibits rich entanglement dynamics, including sudden death and rebirth of entanglement, which are not present in simpler models [157][158]. These dynamics are characterized by parameters that describe the asymmetry and squeezing of the thermal bath, indicating that the environmental state plays a crucial role in determining the outcome [168].

Furthermore, the very definition of entropy and entanglement becomes more nuanced in these complex settings. While entanglement entropy is a useful probe of spacetime architecture in some contexts [91], its calculation in generic states, especially those involving multiple gravitating regions, requires sophisticated tools. Generalized entanglement wedges have been proposed to handle arbitrary gravitating regions, but their application relies on conjectures and is an active area of research [119][120]. In algebraic quantum field theory (AQFT), the algebra of observables for any subregion of spacetime is typically a von Neumann factor of type III1, a property that has profound implications for the structure of entanglement and relative entropy [121][163]. This contrasts sharply with the simpler structure of finite-dimensional Hilbert spaces and means that concepts like the partial trace, which are fundamental to defining entanglement for bipartite systems, must be generalized [149]. Strong mixing properties in KMS states, which describe thermal

equilibrium, are linked to the type III nature of the algebras, further complicating the picture [126]. The modular flow, a concept from Tomita-Takesaki theory, emerges as a key object for characterizing subsystems out of equilibrium and is central to modern approaches in holography and quantum gravity [143][161][164]. These advanced formalisms, while mathematically powerful, highlight the deep conceptual challenges posed by thermal states. They suggest that a naive approach assuming a simple product state is insufficient and that a more rigorous, context-dependent framework is required to make reliable predictions about entanglement generation in a realistic, noisy environment. The failure to properly account for these complexities risks producing results that are artifacts of the model's simplifying assumptions rather than genuine signatures of quantum gravity.

Blocker Category	Description	Key Challenges & Implications
Overwhelming Background Contributions	Environmental noise sources that obscure the target signal ΔScert.	Distinguishing a weak quantum gravity signal from thermal noise, detector imperfections, and other ambient fluctuations [40][141].
Thermal State Entanglement	System-environment entanglement arising from interaction with a thermal bath.	Assumption of a simple product state is invalid; leads to mixed states and complex dynamics [2][149].
Entanglement Breaking Channels	Theoretical possibility that the gravity-mediated interaction destroys entanglement rather than creating it.	Observed effect may be decoherence, not quantum mediation, making it hard to distinguish from other noise sources [7].
Complex Open System Dynamics	Interaction with structured environments (e.g., squeezed thermal baths) leads to non-trivial entanglement evolution.	Requires advanced formalisms like the Schwinger-Keldysh or AQFT; predictions depend heavily on environmental parameters [125][157][158].
Foundational Ambiguities	Deep questions about the nature of entropy, time, and measurement in quantum gravity.	Concepts like the "problem of time" in canonical gravity and the role of observers in quantum theory complicate experimental design [4][55].

In summary, the unfalsifiability claims against a Planck-scale quantum gravity framework are not baseless; they arise from a sober assessment of the immense difficulties in performing high-precision tests in the presence of ubiquitous noise and complex environmental interactions. The challenge is twofold: first, to devise experiments capable of detecting a signal orders of magnitude smaller than the surrounding noise floor, and second, to ensure that the theoretical models used to interpret those experiments accurately reflect the reality of interacting, thermalized quantum systems. Addressing these blockers head-on requires a paradigm shift away from idealized, closed-system thinking toward a more holistic, open-system perspective, integrating insights from quantum metrology, nonequilibrium statistical mechanics, and advanced quantum field theory. The following sections will explore concrete strategies to overcome these challenges, demonstrating that with careful theoretical groundwork and innovative

experimental design, the path toward validating such a profound theory is not entirely blocked.

Advanced Experimental Protocols for Signal Isolation and Enhancement

To counteract the formidable challenges posed by background noise and the complexities of thermal entanglement, a new generation of experimental protocols is required. These protocols must move beyond simple correlation measurements and embrace the full toolkit of modern quantum technologies, focusing on signal enhancement, noise suppression, and the development of robust, unambiguous signatures. The goal is to transform the faint theoretical prediction, ΔScert, into a measurable quantity with a high signal-to-noise ratio (SNR). This involves leveraging quantum resources like entanglement and squeezing, designing clever measurement schemes that serve as entanglement witnesses, and engineering the experimental environment to minimize unwanted interactions. By adopting these advanced strategies, it becomes possible to construct a defense against the criticisms of unfalsifiability, turning the limitations into constraints that guide the design of a more resilient and insightful experiment.

A cornerstone of modern quantum sensing is the use of non-classical states of matter and light to surpass the standard quantum limit (SQL) of precision. Entanglement and, in particular, spin squeezing are indispensable resources for quantum information technology and metrology [14] [137][138]. Squeezed states are a special kind of entangled state that exhibit reduced uncertainty in one observable at the expense of increased uncertainty in another, complementary observable [137]. In the context of atomic or spin systems, this translates to enhanced sensitivity for interferometric sensors, allowing them to achieve precisions that are impossible with uncorrelated (coherent) states [137][138]. For an experiment designed to probe quantum gravity, preparing the component masses or particles in a spin-squeezed state could dramatically amplify the response to the gravitational interaction, effectively boosting the signal ΔScert relative to the intrinsic quantum noise. The generation of spin squeezing has been demonstrated in atomic systems, including Bose-Einstein condensates, showcasing the technological feasibility of creating these resource states [89] [90]. Theoretical proposals for testing quantum gravity already hint at the utility of such states, noting that moderate cooling and the use of squeezed initial states are favorable conditions for observing the relevant effects [139]. By deliberately engineering the initial state to be highly correlated, the experiment can

become more sensitive to the specific dynamical correlations induced by the quantum gravitational mediator, rather than just the total amount of entanglement present.

However, preparing and measuring complex many-body entangled states can be challenging. A more practical and powerful alternative is the use of an "entanglement witness." An entanglement witness is an operator whose expectation value can be measured experimentally. If the expectation value of this operator violates a certain inequality (typically derived under the assumption that the state is separable), then the state is guaranteed to be entangled [174]. Crucially, a strong witness satisfies $W \leq 0$ if and only if the state is entangled, providing a definitive, though not exhaustive, test [174]. This approach circumvents the need for full state tomography, which is exponentially difficult with the number of particles, and focuses on a single, well-defined observable that serves as a smoking-gun signature of the phenomenon in question. Several papers outline prescriptions for constructing such witnesses specifically for the purpose of certifying gravity as a quantum coherent mediator [37] [136]. For example, one proposal describes a scheme based on a Stern-Gerlach apparatus to witness entanglement mediated by gravitational interactions between nano-particles in a spatial superposition [43]. Another outlines a multi-interferometer setup designed to reveal the non-classicality of a gravitational field sourced by a spatially superposed mass [38]. The key advantage of this method is that the witness is constructed to be sensitive to the specific type of entanglement generated by the proposed mechanism, while ideally being insensitive to common decoherence processes. This shifts the experimental goal from measuring a tiny absolute value (ΔScert) to detecting a clear statistical violation of a bound, which is often easier to achieve with high confidence.

Further signal enhancement can be achieved through meticulous environmental engineering and advanced data analysis. A primary strategy for suppressing background noise is to cool the experimental system to ultralow temperatures, approaching the millikelvin or even microkelvin range. This minimizes thermal excitations and reduces the coupling to environmental degrees of freedom. Many cutting-edge experiments in quantum physics rely on this principle, using techniques like laser cooling and evaporative cooling to create Bose-Einstein condensates or ultra-cold atomic gases [89] [95]. Theorists analyzing quantum gravity tests often assume a system prepared in a zero-entropy state or a thermal initial state with moderate cooling, underscoring the necessity of this step [106][139]. Beyond cryogenics, isolating the experiment from external perturbations—such as seismic vibrations, magnetic fields, and electromagnetic radiation—is paramount. This requires sophisticated shielding, vibration isolation platforms, and operation in vacuum chambers.

Even with perfect engineering, residual noise remains. Here, advanced signal processing and characterization techniques become essential. As mentioned, there is ongoing research into methods for characterizing thermal noise in an optical cavity independently of quantum noise, even when the thermal noise amplitude falls below the quantum noise level [140]. Such techniques, which might involve cross-correlation measurements or frequency-domain analysis, could be adapted to separate the quantum gravity signal from other noise sources. Another powerful technique is background subtraction, where measurements are performed under slightly varied conditions to isolate the desired signal. For example, in experiments using atom interferometry, one could measure the phase shift with and without the gravitational interaction enabled (e.g., by turning off a potential that creates the spatial superposition), and subtract the results to cancel common-mode noise [109]. Recent experiments have successfully used collective measurements to directly observe spin-squeezed states with no background subtraction, demonstrating the power of carefully designed measurement schemes to extract the signal cleanly [90].

Finally, the design of the measurement protocol itself can be optimized to maximize the information gained per shot. Instead of relying on a single type of measurement, a multi-parameter approach is often superior. For instance, a multi-interferometer setup can provide multiple, cross-validating pieces of evidence for the non-classicality of the gravitational field [38]. Measuring different correlation functions or different components of the entanglement witness can help distinguish the quantum gravity signal from noise that might mimic only one aspect of the predicted effect. The use of hyperentangled states, where particles are entangled in multiple degrees of freedom simultaneously (e.g., polarization and momentum), offers another avenue for increasing the information density and robustness of the signal [13]. By encoding the quantum gravitational effect into several independent channels, the probability of all channels being simultaneously corrupted by noise is drastically reduced. This multi-pronged approach—combining engineered initial states, entanglement witnesses, environmental control, and sophisticated data analysis—provides a comprehensive strategy for pushing the boundaries of precision and overcoming the noise-related objections to falsifiability. It transforms the problem from one of simply "seeing" a signal to one of designing a measurement protocol that is uniquely tailored to reveal a specific, theoretically motivated signature amidst the noise.

Strategy	Description	Rationale & Benefits
Quantum Metrology Resources	Preparation of particles in non-classical states like spin-squeezed states.	Enhances measurement sensitivity beyond the Standard Quantum Limit, amplifying the weak signal ΔScert relative to intrinsic quantum noise [12] [137] [138].
Entanglement Witnesses	Designing and measuring an operator whose violation of a bound proves entanglement.	Avoids the need for full state tomography; provides a direct, unambiguous signature of the phenomenon without requiring a priori knowledge of the entire state [37] [136] [174].
Environmental Engineering	Cooling systems to ultralow temperatures (mK or μK) and isolating from external perturbations.	Suppresses thermal noise and reduces coupling to environmental degrees of freedom, which are major sources of decoherence and background [89] [95] [139].
Advanced Data Analysis	Characterizing and subtracting known background noise sources, using cross-correlation, etc.	Enables the separation of the faint signal from dominant noise floors, improving the signal-to-noise ratio and statistical significance of the result [90] [140].
Multi-Parameter/Multi-Interferometer Setup	Measuring multiple, independent correlation functions or using several interferometers.	Provides cross-validation of the result; makes it less likely that a single type of noise can mimic the complete set of predicted signatures [13] [38].
Background Subtraction	Performing control measurements and subtracting common-mode noise.	Cancels out systematic noise sources that are present in both signal and background runs, isolating the differential effect of interest [109].

By systematically implementing these strategies, the experimental protocol can be hardened against the primary sources of error and ambiguity. This does not eliminate the fundamental challenges, but it reframes them as engineering problems that can be addressed with existing and near-future technologies. The focus shifts from hoping the signal is large enough to see to actively engineering the system to make the signal large and the noise small. This proactive approach is essential for mounting a credible challenge to the unfalsifiability claims and for paving the way toward a potential empirical confirmation of the quantum gravity framework.

Theoretical Refinements to Overcome Environmental and Initial State Complexities

Addressing the unfalsifiability concern requires not only advanced experimental techniques but also a corresponding sophistication in the underlying theoretical framework. The simplistic assumption of an isolated system starting in a pure, unentangled product state, while pedagogically useful, is a poor approximation for any real-world experiment. To build a robust bridge between theory and experiment, the framework must be refined to naturally accommodate the realities of open quantum systems, thermal environments, and complex initial states. This involves relaxing

restrictive assumptions and embracing more general and powerful mathematical formalisms. By doing so, the theory can generate predictions that are not only more accurate but also more directly comparable to the messy, noisy data produced by actual experiments, thereby strengthening its scientific standing and enhancing its falsifiability.

The most critical refinement is to move beyond the assumption of a simple product initial state. As established, any system in contact with a heat bath will be entangled with it, meaning the combined system-plus-bath state is not a simple tensor product of the system's state and the bath's state [2] [149]. Treating the system in isolation ignores this crucial entanglement and can lead to qualitatively wrong predictions about the generation or preservation of correlations. To properly model such scenarios, one must adopt the language of open quantum systems. The Schwinger-Keldysh, or in-in, formalism is a powerful tool specifically designed for studying the quantum dynamics of systems out of equilibrium, which is precisely the situation when a system is coupled to an environment [125]. This formalism allows for the calculation of expectation values of operators at a given time, correctly accounting for the influence of the environment. It provides a systematic way to compute correlation functions and decoherence rates in the presence of thermal baths with arbitrary spectral properties [24] [125]. Applying this formalism to the proposed quantum gravity experiment would allow for the derivation of equations of motion for the system's density matrix, including terms that describe energy exchange and decoherence due to the interaction with the thermal environment. This would yield a much more realistic prediction for the time-evolved state, from which observables like an entanglement witness could be calculated.

An even more fundamental and powerful approach comes from Algebraic Quantum Field Theory (AQFT). AQFT provides a framework for describing quantum field theories on general curved spacetimes and in arbitrary states, including thermal states [122]. Instead of focusing on a specific representation of the quantum fields, it focuses on the abstract algebra of observables associated with spacetime regions. This perspective is particularly well-suited for addressing the problem of entanglement in thermal states. In AQFT, the thermal state of a system is described by a KMS (Kubo-Martin-Schwinger) state, which encodes the periodicity of correlation functions in imaginary time [63]. The modular operator and modular flow, concepts from Tomita-Takesaki theory, become central objects that characterize the structure of the state on a given algebra [143] [161]. These tools allow for a rigorous definition of entropy, relative entropy, and other thermodynamic quantities for subsystems, even when they are not in a pure state. For instance, recent work has explored how relative entropy behaves between different states in QFT, with implications for defining entanglement entropy in complex situations [105]. Adopting an AQFT-inspired approach would force a more careful treatment of the initial state. Rather

than assuming a product state, one would start with a state that is a solution to the KMS condition, reflecting the thermal nature of the environment. This would lead to predictions that are inherently more robust against the criticisms leveled at simpler models.

Another layer of complexity arises from the fact that the environment itself may not be in a simple thermal state. Realistic environments can be structured, for example, having a spectrum with specific peaks or being in a squeezed state. The interaction of the system with such an environment leads to non-Markovian dynamics, where the future state of the system depends on its entire history, not just its current state. The dynamics of entanglement in a two-qubit system interacting with an initially squeezed thermal bath have been studied in detail, revealing phenomena like entanglement sudden death and revival that are absent in simpler models [157][158]. These studies show that the environmental state is a crucial parameter. A theoretical refinement of the quantum gravity framework must therefore include the environmental state as part of its input. This means the prediction for the observable ΔScert (or the violation of an entanglement witness) would not just be a function of the gravity parameters E*, K, and α_screen, but also of the parameters describing the thermal or squeezed bath. This added complexity is not a drawback; it is a feature that brings the theory closer to reality and opens up new avenues for testing. For example, by tuning the squeezing parameter of the environment, one could perform a controlled experiment to verify whether the predicted decoherence dynamics match the theory's predictions.

Furthermore, the theoretical framework should aim to connect its predictions to established physics in appropriate limits. For example, it should be shown how the proposed Planck-scale framework reduces to the predictions of perturbative quantum gravity at energies far below the Planck scale. Perturbative quantum gravity, despite its non-renormalizability, makes meaningful predictions such as quantum corrections to Newton's law [166]. If the new framework can reproduce these known results, it gains credibility and provides an anchor point for its validity. Similarly, connections to effective field theories, such as deformed conformal field theories that serve as toy models for Planckian physics, could be explored [134]. These theories often exhibit characteristic behaviors, such as a saturation of entanglement entropy, which could provide additional testable signatures [134]. Building these bridges to well-established physics helps to constrain the free parameters (E*, K, α_screen) and provides intermediate steps for validation, making the ultimate Planck-scale test less of a leap into the unknown.

Ultimately, the goal of these theoretical refinements is to produce a set of falsifiable predictions that are explicit, quantitative, and take into account the full complexity of the experimental scenario. This involves: 1. **Deriving the Observable:** Explicitly calculating

the expected value of the chosen observable (e.g., an entanglement witness) as a function of the system's variables, the gravity parameters (E*, K, α_screen), and the environmental parameters (temperature, spectral density, squeezing, etc.). 2.* ***Calculating the Signal-to-Noise Ratio:*** *Comparing the magnitude of the predicted signal to the magnitude of the predicted noise from all relevant sources (thermal, technical, etc.). This calculation is the ultimate arbiter of feasibility. 3.* ***Making Parameterized Predictions:*** *Generating a family of curves or a parameter space plot showing how the predicted signal changes as E*, K, and α_screen are varied. This allows for direct comparison with experimental data, which could then be used to rule out certain regions of the parameter space.

By pursuing this path of theoretical rigor, the framework moves from being a collection of abstract ideas to a predictive model that can be directly confronted with experimental data. This process itself is a form of falsification: if an experiment rules out a wide swath of the parameter space, the theory has been scientifically valuable. Even if the specific Planck-scale hypothesis (E = Ep, |K| = 1/4, α_screen = 1) is ultimately falsified, the refined framework with its free parameters could still provide a viable low-energy effective theory of quantum gravity, contributing significantly to our understanding of the universe. The commitment to confronting the theory with reality, no matter how difficult, is the hallmark of a robust scientific program.

Leveraging Analogue Systems and Established Physics for Intermediate Validation

While direct experimental tests of quantum gravity at the Planck scale remain firmly in the realm of speculation, a pragmatic research strategy involves seeking intermediate validation through two powerful avenues: connecting the framework to established physics in accessible regimes and developing concrete proposals for analogue gravity experiments. These approaches do not prove the theory's high-energy predictions but they ground it in empirically verified domains and provide tangible, near-term pathways to test its core logical tenets. By demonstrating consistency with known phenomena and outlining feasible laboratory experiments, the framework's credibility is significantly enhanced, and the path toward more ambitious tests becomes clearer. This strategy acknowledges the immense difficulty of Planck-scale physics while maintaining a relentless focus on scientific progress, aligning perfectly with the directive to "push forward" and bring the most value to science.

The first step in grounding a radical new theory is to demonstrate its continuity with the vast body of successful physics that came before it. A quantum gravity framework should not exist in a vacuum; it must be able to recover the predictions of General Relativity in the classical limit and those of quantum field theory in the flat-spacetime, low-energy limit. For the framework in question, this means showing how the parameters E*, K, and* α_*screen relate to or reduce to familiar constants and concepts. For example, the theory should be able to reproduce the classical Newtonian gravitational potential and, in a semi-classical approximation, predict corrections to it, similar to how perturbative quantum gravity does* [166]*. The existence of a deformation parameter* α*, representing a ratio of UV to IR limits, suggests a connection to theories with a modified dispersion relation or a fluctuating minimal length, phenomena that are also explored in other approaches like loop quantum gravity (LQG) and doubly special relativity* [6] [135]*. Demonstrating that the framework's predictions for, say, the gravitational interaction between two slowly moving masses reduce to Newton's law, plus calculable quantum corrections dependent on E*, K, and α_screen, would be a major step forward. This provides a foothold in the empirical world, as deviations from Newton's law have been searched for in precision experiments, albeit at much larger length scales than the Planck length [166].

Furthermore, the framework should be connected to the rich phenomenology of quantum field theory in curved spacetime. Concepts like the Unruh effect, Hawking radiation, and entanglement entropy in de Sitter space are well-established results that any viable theory of quantum gravity must accommodate. For instance, the Unruh effect, where an accelerated observer perceives a thermal bath of particles, has been discussed in the context of de Sitter space, where the cosmological horizon gives rise to a temperature [170]. Theories exploring the quantum aspects of gravity often discuss the thermodynamics of black holes, linking entropy to the area of the event horizon [115][171]. If the proposed framework can derive or reproduce these results as special cases, it demonstrates its internal consistency and its capacity to describe known gravitational phenomena. The study of entanglement entropy in de Sitter space, for example, reveals that it is finite and that the curvature can be related to open quantum system effects like Lamb shifts, suggesting deep connections between cosmology, quantum optics, and gravity [99] [100]. By mapping the framework's predictions onto these established areas, one can identify unique, testable signatures that differentiate it from other approaches like LQG or string theory. For example, while LQG predicts a discrete spacetime at the Planck scale [154], the specific form of the deformation parameter α_screen in this framework might lead to different observable consequences in low-energy effective field theories.

The second, and perhaps more exciting, avenue for intermediate validation is the development of analogue gravity experiments. Analogue gravity is the program of using

controllable condensed matter or optical systems to simulate aspects of gravitational physics, such as event horizons, particle creation, and spacetime curvature [36] [93]. While these systems do not probe Planck-scale gravity directly, they provide a laboratory where the fundamental equations governing wave propagation in a dynamic background are mathematically identical to the equations of quantum fields in curved spacetime. This allows for the testing of concepts that are central to quantum gravity in a safe and controllable environment. For the framework under investigation, this means proposing a specific analogue experiment that realizes its key features.

For example, if the framework predicts a specific modification to the speed of propagation for certain modes or a unique pattern of entanglement generation between "phonons" (the analogue of gravitons) in a flowing fluid, one could propose an experiment in a Bose-Einstein condensate or a nonlinear optical medium to search for this effect. The spin-2 mode of a chiral graviton theory has been argued to be analogous to fluctuations of a tensor order parameter in condensed matter systems [36]. A systematic development of an effective theory for such a massive chiral graviton could lead to proposals for experiments in fractional quantum Hall states or other topological materials [28]. Molecular polaritons, which are quasiparticles formed from the hybridization of molecular excitations and photonic modes, offer another platform for simulating quantum electrodynamics and could be adapted to explore quantum gravity-inspired models [70] [147]. Digital quantum simulators, using platforms like trapped ions or superconducting qubits, could also be programmed to mimic the dynamics of quantum fields in a curved spacetime background, providing a clean testbed for theoretical predictions [35].

Developing such an analogue proposal would require translating the abstract parameters of the quantum gravity framework (E^*, K, α_{screen}) into concrete, tunable parameters of the condensed matter or optical system (e.g., sound speed, nonlinearity coefficient, lattice spacing). Success in this endeavor would constitute a major validation. Observing a phenomenon in the analogue system that matches the theory's prediction would provide strong evidence for the underlying logic of the framework, even if it doesn't confirm the specific Planck-scale values of the parameters. It would show that the framework is not just a mathematical curiosity but a physically meaningful model that can make falsifiable predictions about wave phenomena in general. This approach has a long history of success in physics; for example, the observation of Hawking-like radiation in sonic black holes in Bose-Einstein condensates, while indirect, provided a powerful proof-of-concept for the entire analogue gravity program [101].

Connecting to established physics and building analogue models serve as crucial stepping stones. They transform the theory from a purely high-energy hypothesis into a broader physical framework with testable consequences across different energy scales. The process forces the theorists to be more concrete, to confront the link between their abstract equations and real-world observables. Each successful connection or analogue prediction builds a case for the framework's plausibility and refines our understanding of its implications. This incremental approach is the most reliable path toward eventually tackling the ultimate, and most difficult, challenge of directly testing quantum gravity at its native energy scale.

Validation Strategy	Description	Rationale & Benefits
Connection to Low-Energy Physics	Demonstrating that the framework reproduces known physics like Newton's law, GR, and QFT in appropriate limits.	Anchors the theory in empirically verified domains; provides intermediate, testable predictions (e.g., corrections to Newton's law) [166].
Consistency with QFT in Curved Spacetime	Showing that the framework can derive or accommodate established results like Hawking/Unruh effects and entanglement entropy.	Proves internal consistency with a well-understood regime of quantum gravity; links the theory to cosmology and black hole physics [99] [115] [170].
Proposal of Analogue Gravity Experiments	Designing a laboratory experiment in a condensed matter or optical system that simulates the key predictions of the framework.	Provides a tangible, near-term pathway to test the theory's core logic in a controllable environment; bypasses the need for Planck-scale energies [28] [35] [36].
Translation to Analogue Parameters	Mapping the abstract gravity parameters (E*, K, α_screen) to tunable parameters in the analogue system (e.g., sound speed, nonlinearity).	Makes the theory's predictions concrete and experimentally accessible; allows for systematic exploration of the parameter space in the lab.
Exploration of Effective Field Theories	Connecting the framework to effective theories like deformed CFTs or considering T. ¯. T deformations.	Links the UV-complete theory to low-energy observables; may reveal new, testable signatures like saturated entanglement entropy [117] [134].

By pursuing these intermediate goals, the research program avoids the pitfall of becoming a purely speculative exercise. It maintains a continuous dialogue with empirical science, using established results as both a guide and a check. This strategy not only enhances the credibility and falsifiability of the framework but also ensures that every theoretical advance contributes directly to a long-term vision of experimental verification, however distant that vision may seem today.

Synthesis and Actionable Roadmap for Scientific Validation

The comprehensive examination of theoretical and experimental concerns related to the unfalsifiability of a Planck-scale quantum gravity framework reveals that while the

challenges are immense, they are not insurmountable. The primary obstacles—overwhelming background noise and the complexities of entanglement in thermal states—are not fatal flaws but rather represent the frontier of contemporary experimental physics and quantum theory. The unfalsifiability claims stem from a legitimate concern about the feasibility of isolating a minuscule signal from a noisy environment, but they do not preclude the possibility of a solution. By adopting a multi-pronged strategy that combines theoretical refinement, advanced experimental protocols, and intermediate validation through established physics and analogue systems, it is possible to construct a credible path toward scientific validation. This final section synthesizes the findings of this report into a concrete, actionable roadmap designed to dismantle the unfalsifiability barrier and maximize the scientific contribution of the framework.

The core insight of this analysis is that the strategy must be adaptive and layered. We cannot simply wish the challenges away; instead, we must systematically deconstruct them and build defenses against each one. The roadmap begins with a deep theoretical engagement to ensure the framework is not only internally consistent but also generates explicit, falsifiable predictions. This involves moving beyond idealized assumptions and employing rigorous formalisms like the Schwinger-Keldysh and Algebraic Quantum Field Theory to model the system's interaction with a realistic, structured, and thermalized environment. The next step is to leverage the full arsenal of quantum technologies to design experiments that are not just sensitive but are specifically tailored to amplify the target signal while suppressing noise. Finally, this effort must be anchored in empirical reality by connecting the theory to known physics and by charting a course for tangible, near-term tests via analogue gravity experiments.

Based on this synthesis, the following actionable roadmap is proposed:

Phase 1: Foundational Theoretical Development and Signal Prediction

The immediate priority is to move from an abstract framework to a set of concrete, quantitative predictions. This phase involves: 1. **Derive the Core Observable:** Within the specified quantum gravity framework, perform a detailed calculation to derive the explicit form of the predicted observable, ΔScert, or, more practically, the expectation value of a proposed entanglement witness. This calculation must be done for a system interacting with a non-trivial environment, such as a squeezed thermal bath, to reflect realistic conditions [157][158]. 2. **Establish the Signal-to-Noise Ratio (SNR):** Quantify the predicted signal strength and compare it directly to the predicted noise levels from all relevant sources, including thermal noise, detector dark counts, and technical noise [40][141]. This SNR calculation is the most critical output of this phase, as it will definitively determine the feasibility of the experiment. 3. **Parameterize the Predictions:** Generate a

comprehensive set of predictions as a function of the free parameters E*, K, and α_screen. This will create a "prediction map" that can be directly compared against future experimental data, allowing for the exclusion of entire regions of the theory's parameter space [136].

Phase 2: Advanced Experimental Protocol Design and Optimization

If Phase 1 yields a promising, albeit challenging, SNR, the focus shifts to experimental design. This phase aims to engineer the maximum possible enhancement of the signal and suppression of the noise. 1. **Select and Optimize the Physical Platform:** Choose a suitable physical system for the experiment, such as ultracold atoms in an optical lattice, trapped ions, or superconducting qubits [89] [95]. The selection should be guided by the parameters needed for the optimal SNR calculation from Phase 1 (e.g., required coherence times, interaction strengths). 2. **Implement Quantum Metrology Techniques:** Design the protocol to begin with a highly entangled initial state, such as a spin-squeezed state, to exploit the enhanced sensitivity offered by quantum metrology [137] [138]. Simultaneously, optimize the measurement basis to correspond to the expectation value of the chosen entanglement witness, maximizing the contrast of the signal [37]. 3. **Engineer the Environment:** Implement active cooling to reach the lowest possible temperature, minimizing thermal noise contributions [139]. Employ sophisticated shielding and vibration isolation to reduce technical noise. Explore the use of structured reservoirs or tailored spectral densities to either suppress specific noise channels or, conversely, to use them as a tool to probe the system's response [24].

Phase 3: Intermediate Validation and Long-Term Vision

Regardless of the outcome of the direct test, this phase ensures the continued scientific relevance and contribution of the research program. 1. **Connect to Low-Energy Physics:** Systematically demonstrate how the framework reduces to known physics in appropriate limits. Derive quantum corrections to Newton's law or analyze its predictions for the behavior of quantum fields in weak gravitational potentials to show consistency with established results [166]. 2. **Develop an Analogue Gravity Proposal:** Translate the core principles of the quantum gravity framework into a proposal for a feasible analogue experiment. Identify a condensed matter or optical system where the key features of the theory (e.g., a specific dispersion relation, modified commutation relations) can be simulated [28] [36]. The goal is to find a laboratory-scale system where the theory's unique predictions can be tested without reaching Planck-scale energies. 3. **Explore Alternative Signatures:** If the primary signature ΔScert proves too elusive, investigate alternative, potentially stronger observables. This could include higher-order correlation functions,

specific patterns of entanglement entropy scaling, or signatures in the system's response to external driving forces [49] [91].

In conclusion, the journey to validate a Planck-scale quantum gravity framework is a marathon, not a sprint, and it requires navigating a landscape filled with profound theoretical and daunting experimental challenges. The unfalsifiability claims are a reflection of this difficulty, but they are not a verdict. By confronting the blockers of background noise and thermal entanglement head-on, and by pursuing a disciplined, multi-phase strategy of theoretical prediction, experimental optimization, and intermediate validation, it is possible to transform the framework from a subject of philosophical debate into a topic of empirical science. The proposed roadmap provides a clear and adaptable path forward. It demands rigor, creativity, and persistence, but it also promises the immense reward of either confirming a deep truth about the fabric of spacetime or, at the very least, refining our understanding of the physical laws that govern our universe in a decisive and scientifically rigorous manner.

Reference

1. Testing the Minimum System Entropy and the Quantum ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10670145/
2. Why do we need mixed states in quantum mechanics? https://physics.stackexchange.com/questions/733025/why-do-we-need-mixed-states-in-quantum-mechanics
3. A quantum microtubule substrate of consciousness is ... https://academic.oup.com/nc/article/2025/1/niaf011/8127081
4. Quantum theory cannot consistently describe the use of itself https://www.nature.com/articles/s41467-018-05739-8
5. The Theory of Derived Probability and Entanglement ... https://zenodo.org/records/17189570/files/20250924_Theory_of_Derived_Probability_and_Entanglement_Compression_v7.pdf?download=1
6. quantum theory at planck scale, limiting values, deformed gravity ... https://www.researchgate.net/publication/49943345_QUANTUM_THEORY_AT_PLANCK_SCALE_LIMITING_VALUES_DEFORMED_GRAVITY_AND_DARK_ENERGY_PROBLEM

7. Can gravity mediate the transmission of quantum ... https://arxiv.org/html/2504.05998v3
8. Entanglement in gravity tests may not be as clear-cut ... https://www.linkedin.com/posts/phys-org_using-entanglement-to-test-whether-gravity-activity-7387450884823085056--CWV
9. Quantum gravitational decoherence from fluctuating minimal ... https://pmc.ncbi.nlm.nih.gov/articles/PMC8298405/
10. [PDF] Entanglement Witnesses of Condensation for Enhanced Quantum ... https://arxiv.org/pdf/2511.17749
11. Witnessing Entanglement and Quantum Correlations in Condensed ... https://advanced.onlinelibrary.wiley.com/doi/10.1002/qute.202400196
12. Quantum metrology with a continuous-variable system - IOPscience https://iopscience.iop.org/article/10.1088/1361-6633/ae00d8
13. Entanglement source preparation via three-photon quantum ... https://www.researchgate.net/publication/397386193_Entanglement_source_preparation_via_three-photon_quantum_scissors_enhanced_by_realistic_conditions
14. Maximal entanglement increase with single-photon subtraction https://www.semanticscholar.org/paper/Maximal-entanglement-increase-with-single-photon-Zhang-Jing/140e03d923b9bff7b07b0b7d053db770ddfb8be8
15. Why Do We Want a Theory of Quantum Gravity? - arXiv https://arxiv.org/html/2505.04858v1
16. (PDF) A Unified Framework for Gravity and Quantum Structure https://www.researchgate.net/publication/392081903_A_Unified_Framework_for_Gravity_and_Quantum_Structure
17. Issue 2 - Volume 43 - Chinese Physics Letters - IOPscience https://iopscience.iop.org/issue/0256-307X/43/2
18. Generalized Entanglement Wedges from the Gravitational ... https://arxiv.org/pdf/2506.10064
19. Solvable models of quantum black holes: a review on Jackiw ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10390458/
20. Simulating Open Quantum System Dynamics on NISQ ... https://pubs.acs.org/doi/10.1021/acs.jctc.3c00316
21. Manuscript_fb470dc75808ae737... https://www.sciencedirect.com/science/article/am/pii/S0003491622001336
22. The irreversible relaxation of inflation https://www.sciencedirect.com/science/article/pii/S0370269325002801

23. arXiv:2104.08881v4 [gr-qc] 9 Sep 2022 https://arxiv.org/pdf/2104.08881
24. Nonequilibrium quantum heat transport between structured ... https://iopscience.iop.org/article/10.1088/1367-2630/ad5bfb
25. Simulating Quantum Vibronic Dynamics at Finite ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC7831969/
26. Heating and cooling are fundamentally asymmetric and ... https://www.nature.com/articles/s41567-023-02269-z
27. Structure-Fair Quantum Circuit Complexity: An Auditable ... https://arxiv.org/html/2509.18205v2
28. Chiral Graviton Theory of Fractional Quantum Hall States https://arxiv.org/pdf/2509.04408
29. Dynamics in canonical models of loop quantum gravity https://arxiv.org/pdf/1910.00469
30. Towards Worst-Case Guarantees with Scale-Aware ... https://arxiv.org/html/2602.05184v1
31. arXiv:2307.13416v1 [gr-qc] 25 Jul 2023 https://arxiv.org/pdf/2307.13416
32. Forbidden neutrinogenesis https://arxiv.org/html/2412.18278v2
33. Fundamental Physics in 2025: Status, Decisive Targets, ... https://arxiv.org/html/2512.21445v1
34. A Scenario for Strong Gravity without Extra Dimensions https://arxiv.org/pdf/hep-th/0602183
35. Digital quantum simulation of cosmological particle ... https://arxiv.org/html/2410.02412v2
36. Chiral Graviton Theory of Fractional Quantum Hall States https://arxiv.org/html/2509.04408v1
37. Entanglement witness for combined atom interferometer ... https://arxiv.org/html/2503.09990v1
38. Testing Whether Gravity Acts as a Quantum Entity When ... https://arxiv.org/html/2307.08133v5
39. Sudden Decoherence by Resonant Particle Excitation for ... https://arxiv.org/html/2501.18147v1
40. arXiv:2101.08086v1 [quant-ph] 20 Jan 2021 https://arxiv.org/pdf/2101.08086
41. Analogue many-body gravitating quantum systems with a ... https://arxiv.org/html/2602.23319v1
42. Entanglement dynamics of photon pairs and quantum ... https://arxiv.org/html/2209.02099v3

43. arXiv:2304.14638v1 [quant-ph] 28 Apr 2023 https://arxiv.org/pdf/2304.14638
44. Gravity mediated entanglement between light beams as a ... https://arxiv.org/html/2210.12713v2
45. arXiv:2110.14695v1 [quant-ph] 27 Oct 2021 https://arxiv.org/pdf/2110.14695
46. arXiv:2101.11629v3 [quant-ph] 26 Jan 2022 https://arxiv.org/pdf/2101.11629
47. State-Selective Signatures of Quantum and Classical ... https://arxiv.org/html/2603.05731v1
48. State-Selective Signatures of Quantum and Classical ... https://arxiv.org/pdf/2603.05731
49. Radius-Flow Entanglement in Hadron States and ... https://arxiv.org/html/2603.03064v1
50. Perspectives in Quantum Gravity https://arxiv.org/html/2412.08696v2
51. Quantum Gravity as Gravitized Quantum Theory https://arxiv.org/pdf/2407.06207
52. Thermodynamic Gravity with Non-Extensive Horizon ... https://arxiv.org/html/2602.20430v1
53. Cosmological Implications of Thermodynamic Split ... https://arxiv.org/pdf/2510.10441
54. arXiv:1909.06340v3 [quant-ph] 20 Jul 2020 https://arxiv.org/pdf/1909.06340
55. Emergent spacetime from spatial energy potentiality https://arxiv.org/html/2502.18524v5
56. The Cellular Automaton Interpretation of Quantum Mechanics https://arxiv.org/pdf/1405.1548
57. arXiv:2301.08699v1 [hep-ex] 20 Jan 2023 https://arxiv.org/pdf/2301.08699
58. Science Spoofs, Physics Pranks and Astronomical Antics - ar5iv https://ar5iv.labs.arxiv.org/html/2103.17057
59. PHYSICISTS ARE STILL JOKING Almanac of Physics and ... https://arxiv.org/html/2512.17948v2
60. arXiv:2103.17057v2 [physics.hist-ph] 27 May 2021 https://arxiv.org/pdf/2103.17057
61. Machine Learning Feb 2025 https://www.arxiv.org/list/cs.LG/2025-02?skip=2050&show=2000
62. Computer Science Apr 2025 https://www.arxiv.org/list/cs/2025-04?skip=9350&show=2000
63. arXiv:2402.03939v2 [hep-th] 13 Nov 2024 https://arxiv.org/pdf/2402.03939
64. Evaporation of black holes in flat space entangled with an ... https://academic.oup.com/ptep/article/2022/1/013B13/6461111

65. Quantum computational complexity from quantum information ... https://link.springer.com/article/10.1140/epjc/s10052-022-10037-1
66. Thermal Quantum Correlations in Two Gravitational Cat ... https://www.mdpi.com/2218-1997/9/2/71
67. A statistical mechanics framework for constructing ... https://academic.oup.com/pnasnexus/article/2/12/pgad417/7473625
68. Abstract https://arxiv.org/html/2603.15585v1
69. Complex quantum networks: a topical review https://iopscience.iop.org/article/10.1088/1751-8121/ad41a6
70. Understanding Polaritonic Chemistry from Ab Initio Quantum ... https://pubs.acs.org/doi/10.1021/acs.chemrev.2c00788
71. Melting Is Well-Known, but Is It Also Well-Understood? - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10722469/
72. Convective quasi − equilibrium - Yano - 2012 - AGU Journals https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2011RG000378
73. Quantum Gravity at the Planck Scale A Critical Review of Current ... https://www.researchgate.net/publication/395915657_Quantum_Gravity_at_the_Planck_Scale_A_Critical_Review_of_Current_Approaches_String_Theory_Loop_Quantum_Gravity_and_Beyond
74. Thermal state entanglement entropy on a quantum graph - arXiv https://arxiv.org/abs/1909.00449
75. Thermal state entanglement entropy on a quantum graph https://www.researchgate.net/publication/335599254_Thermal_state_entanglement_entropy_on_a_quantum_graph
76. Abstract - arXiv.org https://arxiv.org/html/2412.08696v3
77. Testing the uniqueness of a unified theory based on topology and ... https://www.researchgate.net/publication/389673692_Testing_the_uniqueness_of_a_unified_theory_based_on_topology_and_geometry
78. [PDF] Quantum Gravity Emergence from Entanglement in a Multi ... - HAL https://hal.science/hal-02881693v1/file/epr_paper_6_9_20_preprint.v1_distr_1.pdf
79. Singularities and Black Holes - Stanford Encyclopedia of Philosophy https://plato.stanford.edu/archives/fall2025/entries/spacetime-singularities/
80. Quantum foundations for quantum technologies in the ... https://arxiv.org/html/2603.04630v1
81. Quantum foundations for quantum technologies in the ... https://arxiv.org/pdf/2603.04630

82. Quantum Entanglement in High Energy Physics https://intech-files.s3.amazonaws.com/a043Y00000zXAyYQAW/0015109_Authors_Book%20%282024-05-09%2013%3A59%3A36%29.pdf
83. Reaching relativistic distances and velocities https://www.researchgate.net/publication/227716852_Fundamental_quantum_optics_experiments_conceivable_with_satellites_-_Reaching_relativistic_distances_and_velocities
84. The simple reason why classical gravity can entangle https://www.researchgate.net/publication/397280426_The_simple_reason_why_classical_gravity_can_entangle
85. Visions in Quantum Gravity https://arxiv.org/pdf/2412.08696
86. Cosmological Implications of Thermodynamic Split ... https://arxiv.org/html/2510.10441v1
87. Extending the QMM Framework to the Strong and Weak ... https://arxiv.org/pdf/2504.03817
88. Generalized Symmetry Breaking Scales and Weak Gravity ... https://arxiv.org/pdf/2202.05866
89. [1203.5359] Spin squeezing, entanglement and quantum ... https://arxiv.org/abs/1203.5359
90. Detection of a quantum phase with entanglement ... https://www.researchgate.net/figure/Detection-of-a-quantum-phase-with-entanglement-enhanced-sensitivity-We-apply-a-small_fig3_257657125
91. [PDF] Observations on entanglement entropy in massive QFT's https://www.semanticscholar.org/paper/b279d5fcf0a5ce46f3a70280343fd28c812db22c
92. Asymptotically safe quantum gravity: functional and lattice ... - arXiv https://arxiv.org/html/2509.26352v1
93. Atomic Quantum Technologies for Quantum Matter and ... - MDPI https://www.mdpi.com/2227-7080/12/5/64
94. (PDF) Relational Time Flow from Entanglement Density https://www.researchgate.net/publication/394219643_Relational_Time_Flow_from_Entanglement_Density_Theory_Parameter-Free_Checks_and_Experimental_Protocols
95. [PDF] Coherence and relaxation of an optically-driven bosonic quantum gas https://theses.hal.science/tel-02932168v1/file/these_bosch_aguilera_manel_2019.pdf
96. Computer Science https://arxiv.org/list/cs/new
97. Finite Entanglement Entropy in Asymptotically Safe Quantum Gravity https://arxiv.org/abs/1804.02162

98. Entanglement generation between two comoving Unruh-DeWitt ... https://arxiv.org/html/2404.11931v1
99. Entanglement entropy in de Sitter space - ResearchGate https://www.researchgate.net/publication/232718089_Entanglement_entropy_in_de_Sitter_space
100. Relating the curvature of De Sitter universe to open quantum Lamb ... https://link.springer.com/article/10.1140/epjc/s10052-021-08977-1
101. [PDF] Entanglement and decoherence in cosmology and in analogue ... https://theses.hal.science/tel-04260012v1/file/111254_MICHELI_2023_archivage.pdf
102. [PDF] Non-equilibrium Structures and Cosmic Evolution in Gravitational ... https://zenodo.org/records/18940546/files/3-Daisuke_SATO-ORCID0009-0008-3878-4169_0307_1660.pdf?download=1
103. 1 Introduction - arXiv.org https://arxiv.org/html/2404.13684v1
104. Aspects of Generalized Entropy And Quantum Null Energy Condition https://escholarship.org/uc/item/6qt4k7rz
105. Gravitational Algebras with Two Areas | Request PDF - ResearchGate https://www.researchgate.net/publication/398357399_Gravitational_Algebras_with_Two_Areas
106. [PDF] arXiv:2502.01933v1 [hep-th] 4 Feb 2025 https://arxiv.org/pdf/2502.01933
107. Testing the quantum nature of gravity through interferometry - arXiv.org https://arxiv.org/html/2506.13085v1
108. Quantum Gravity Tests Coming Soon - ResearchGate https://www.researchgate.net/publication/400929174_Quantum_Gravity_Tests_Coming_Soon
109. First Feasible Tabletop Test of Quantum Gravity via Gravitational ... https://www.academia.edu/144661347/First_Feasible_Tabletop_Test_of_Quantum_Gravity_via_Gravitational_Aharonov_Bohm_Effect_Updated_
110. Entropy, Volume 27, Issue 7 (July 2025) – 117 articles - MDPI https://www.mdpi.com/1099-4300/27/7
111. Arxiv今日论文| 2026-03-04 - 闲记算法 http://lonepatient.top/2026/03/04/arxiv_papers_2026-03-04.html
112. (PDF) Quantum Semantic Physics From QSIT Theory to Empirical ... https://www.researchgate.net/publication/397755773_Quantum_Semantic_Physics_From_QSIT_Theory_to_Empirical_Validation_of_the_Resonant_Mode_in_Multi-Agent_Systems
113. LESU integrated light group sound group assembly CNC shell ... https://www.aliexpress.com/item/1005009594007071.html

114. [XLS] China https://www.nsfc.gov.cn/Portals/0/fj/fj20230220_01.xlsx

115. [PDF] Entanglement entropy and correlations in loop quantum gravity https://www.semanticscholar.org/paper/7a4b2e87b7b4426f6a1f7631bfda731b305c8ea9

116. Holographic derivation of entanglement entropy from AdS/CFT https://www.researchgate.net/publication/311962958_Holographic_derivation_of_entanglement_entropy_from_AdSCFT

117. [PDF] arXiv:2503.09997v2 [hep-th] 12 May 2025 https://arxiv.org/pdf/2503.09997

118. A note on quantum supergravity and AdS/CFT - Semantic Scholar https://www.semanticscholar.org/paper/A-note-on-quantum-supergravity-and-AdS%2FCFT-Bodendorfer/334e1291df135461c7cd6c24359598536059ba63

119. Hollow-grams: Generalized Entanglement Wedges from ... https://arxiv.org/abs/2506.10064

120. [2208.04993] Entanglement Wedges for Gravitating Regions https://arxiv.org/abs/2208.04993

121. (PDF) Algebras for generalized entanglement wedges https://www.researchgate.net/publication/398134583_Algebras_for_generalized_entanglement_wedges

122. Quantum information in quantum field theory and ... https://arxiv.org/pdf/2203.07117

123. On the Dressed Photon Constant and Its Implication for a ... https://www.mdpi.com/2073-8994/13/4/593

124. Asymptotically isometric codes for holography https://arxiv.org/pdf/2211.12439

125. Schwinger-Keldysh formalism I: BRST symmetries and ... https://arxiv.org/pdf/1610.01940

126. JHEP01(2024)112 - Springer https://link.springer.com/content/pdf/10.1007/JHEP01(2024)112.pdf

127. [PDF] arXiv:2306.07342v1 [quant-ph] 12 Jun 2023 https://arxiv.org/pdf/2306.07342

128. Open-system dynamics of entanglement:a key issues review https://iopscience.iop.org/0034-4885/78/4/042001/pdf/0034-4885_78_4_042001.pdf

129. [PDF] Many-body open quantum systems - HAL https://hal.science/hal-05408438v1/file/SciPostPhysLectNotes_99.pdf

130. [PDF] Quantum Information and Gravity: A Window by Newton Cheng https://escholarship.org/content/qt0b61k2m8/qt0b61k2m8_noSplash_aa25fe1d6011bbb0a3f47f3684de31ea.pdf?t=s3e154

131. Asymptotic safety, quantum gravity, and the swampland https://arxiv.org/html/2502.12290v3

132. [PDF] Aspects of quantum gravity in AdS3 /CFT2 - IAEA International ... https://inis.iaea.org/records/ybs0c-cxh92/files/50067514.pdf?download=1

133. (PDF) The Physics of Gravity and Anti-Gravity in Relation with String ... https://www.researchgate.net/publication/396423521_The_Physics_of_Gravity_and_Anti-Gravity_in_Relation_with_String-and_Loop_Quantum_Gravity_Theories_the_Requirement_of_Cosmic_Consciousness_Human-and_Artificial_Intelligence

134. deformed conformal field theories: A toy model for Planckian physics https://arxiv.org/html/2507.16262v2

135. pc-Gravity as a geometric resolution of the black hole information ... https://iopscience.iop.org/article/10.1088/1402-4896/adfd9b

136. A Spin Entanglement Witness for Quantum Gravity - arXiv https://arxiv.org/abs/1707.06050

137. (PDF) Spin squeezing, entanglement and quantum metrology with ... https://www.researchgate.net/publication/221941000_Spin_squeezing_entanglement_and_quantum_metrology_with_Bose-Einsteincondensates

138. Spin squeezing, entanglement and quantum metrology with Bose ... https://www.semanticscholar.org/paper/Spin-squeezing%2C-entanglement-and-quantum-metrology-Gross/0aeb2aa4f393145d048d96767e37b512b9bcc144

139. A Spin Entanglement Witness for Quantum Gravity | Request PDF https://www.researchgate.net/publication/318560350_A_Spin_Entanglement_Witness_for_Quantum_Gravity

140. Thermal Noise Measurement Below the Standard Quantum ... https://arxiv.org/html/2507.02196v1

141. (PDF) The Hannover thermal noise experiment https://www.researchgate.net/publication/231129591_The_Hannover_thermal_noise_experiment/download

142. Nonperturbative gravity corrections to bulk reconstruction https://iopscience.iop.org/article/10.1088/1751-8121/acef7d

143. JHEP08(2025)166 - Springer Nature https://link.springer.com/content/pdf/10.1007/JHEP08(2025)166.pdf

144. Quantum logic and entanglement by neutral Rydberg atoms https://iopscience.iop.org/article/10.1088/2058-9565/ac18b8

145. [PDF] arXiv:2411.07997v2 [cond-mat.quant-gas] 9 May 2025 https://arxiv.org/pdf/2411.07997

146. [PDF] Channel Mixing and Spin transport in the Integer Quantum Hall Effect https://theses.hal.science/tel-00680044/file/21012_VENTURELLI_2011_archivage_1_.pdf

147. Molecular Polaritons for Chemistry, Photonics and Quantum ... https://pubs.acs.org/doi/10.1021/acs.chemrev.3c00662

148. [PDF] University of Groningen Entangled Systems Schut, Martine - CORE https://core.ac.uk/download/686435789.pdf

149. How to discern that a subsystem of entangled system is in a mixed ... https://physics.stackexchange.com/questions/207402/how-to-discern-that-a-subsystem-of-entangled-system-is-in-a-mixed-state

150. An operational distinction between quantum entanglement ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC11667580/

151. [PDF] Diagnosing Metal-Insulator and Hawking-Page Transitions - arXiv https://arxiv.org/pdf/2601.00071

152. Spontaneous irreversibility and objective thermalization in ... https://arxiv.org/pdf/2504.16197

153. [PDF] Asymptotic safety, quantum gravity, and the swampland - arXiv.org https://arxiv.org/pdf/2502.12290

154. [PDF] Black Holes in Loop Quantum Gravity - HAL https://hal.science/hal-01645217v1/document

155. Hilbert Bundles and Holographic Space–Time Models - MDPI https://www.mdpi.com/2674-0346/4/2/7

156. Entanglement and spin squeezing in non-Hermitian phase transitions https://pubmed.ncbi.nlm.nih.gov/25554863/

157. Entanglement Dynamics in Two-Qubit Open System Interacting with ... https://arxiv.org/abs/0810.5322

158. [PDF] Entanglement dynamics in two-qubit open system interacting ... https://www.semanticscholar.org/paper/cebcb6e5810f1921ff76834cf90c0a56cbcc8bfc

159. The only restriction on the initial state is that it be unentangled... https://www.researchgate.net/figure/The-only-restriction-on-the-initial-state-is-that-it-be-unentangled-ie-a-product-state_fig1_237000206

160. (PDF) Planck Scale Kinematics and the Pierre Auger Observatory https://www.academia.edu/17845991/Planck_Scale_Kinematics_and_the_Pierre_Auger_Observatory

161. An intuitive construction of modular flow https://arxiv.org/pdf/2309.16766

162. [PDF] arXiv:1308.4348v2 [gr-qc] 15 Feb 2014 https://arxiv.org/pdf/1308.4348

163. Crossed product algebras and generalized entropy for ... https://arxiv.org/pdf/2306.07323

164. JHEP06(2025)242 - Springer https://link.springer.com/content/pdf/10.1007/JHEP06(2025)242.pdf

165. arXiv:2301.08753v2 [hep-th] 13 May 2024 https://arxiv.org/pdf/2301.08753
166. [PDF] The Many Facets of Black Hole Physics in AdS/CFT - eScholarship https://escholarship.org/content/qt99d7b4wm/qt99d7b4wm.pdf
167. Review Articles 1991-2023 - Classical and Quantum Gravity https://iopscience.iop.org/journal/1361-6382/page/Review-Articles-1991-2023
168. Entanglement bounds for squeezed non-symmetric thermal ... https://arxiv.org/abs/quant-ph/0309156
169. Entanglement generation between Unruh-DeWitt detectors ... https://arxiv.org/abs/2404.11931
170. [1011.3336] On the Unruh effect in de Sitter space https://arxiv.org/abs/1011.3336
171. [2207.14274] Self-Gravity and Bekenstein-Hawking Entropy https://arxiv.org/abs/2207.14274
172. [PDF] 1996 年版物理学名词修订英中对照初稿本 https://wulixb.iphy.ac.cn/fileWLXB/cms/news/info/upload/0b531a2f-aaaa-465e-9eac-3e33edbdb2c6.pdf
173. Entanglement between more than two hundred macroscopic atomic ... https://pmc.ncbi.nlm.nih.gov/articles/PMC5640660/
174. [PDF] Quantum Metrology, Imaging, and Communication https://qsip.sjtu.edu.cn/__local/0/F8/0E/A3732BFCE4FE1701E476A7DC6F1_DC98E06B_61BBFE.pdf